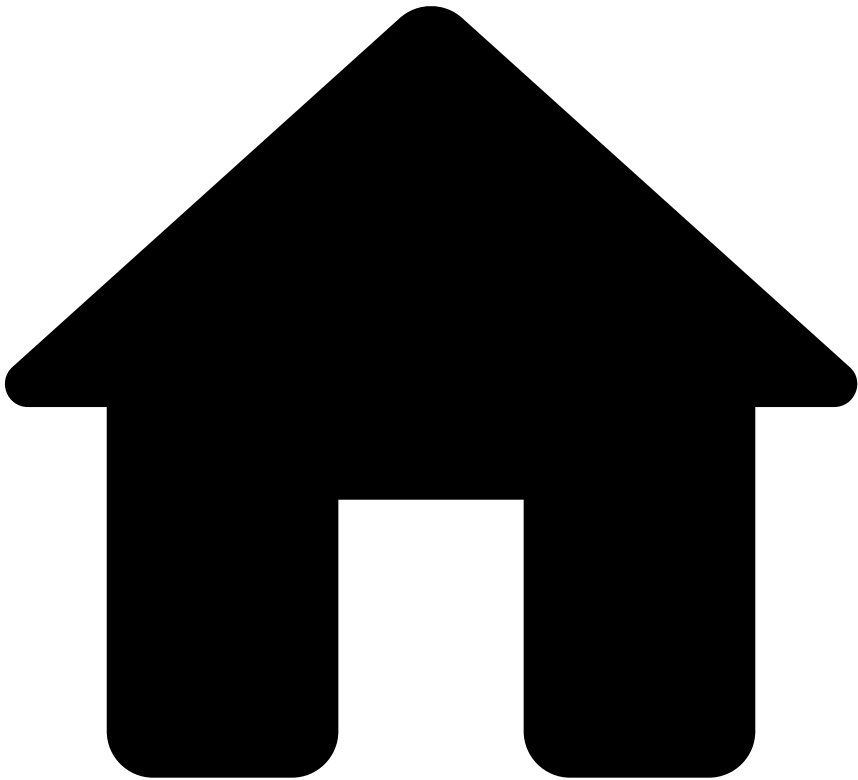


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Operators in Rules

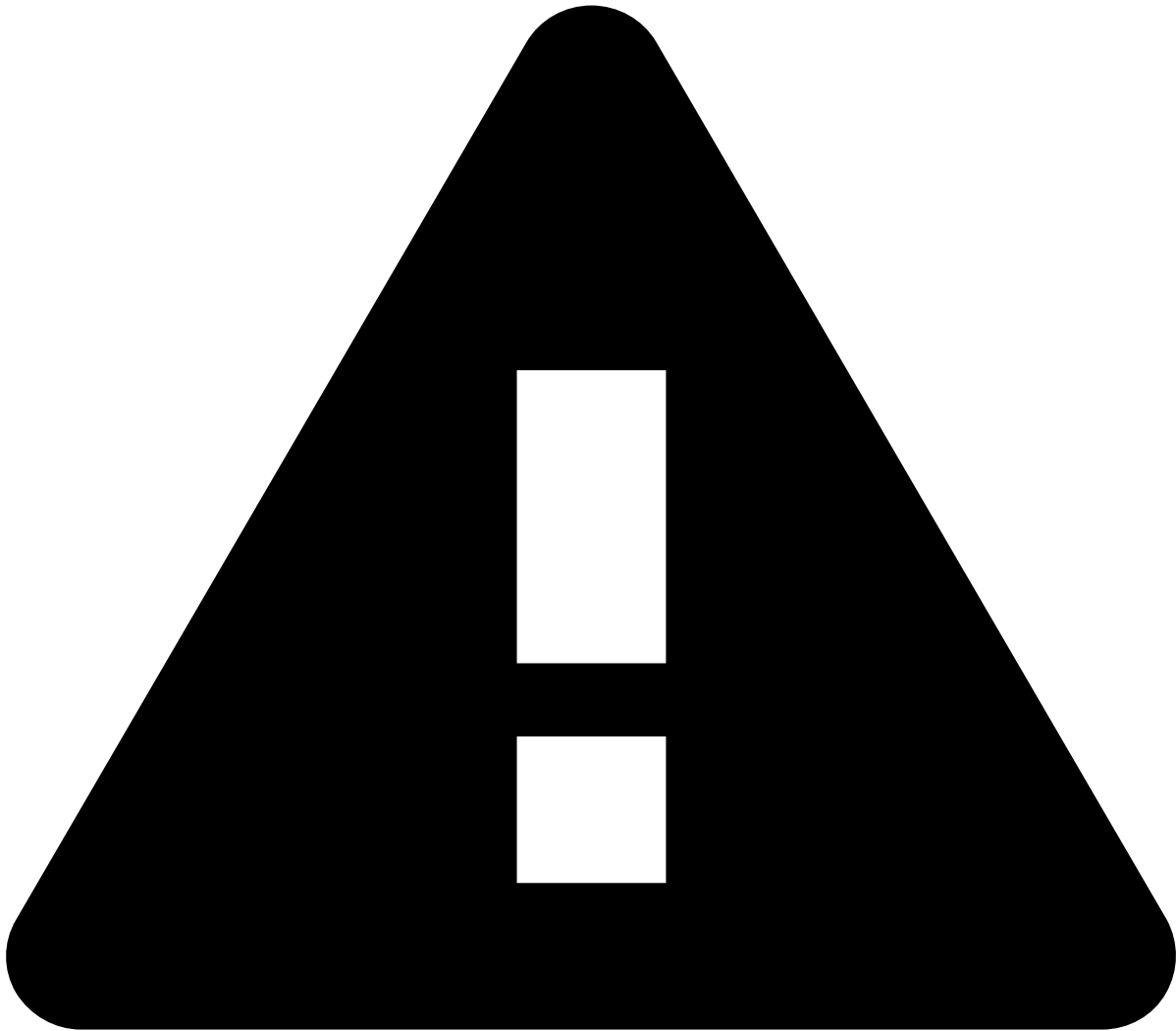
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The Rule Definition Language (RDL) is designed for defining variables and conditions for Rules in Pulse. RDL queries event data, and transforms it using operators and function calls.

This topic gives you a list of operators that you can use when writing rules.

Operators

The following table explains the meaning of each supported operator.



Rules failing with null values

Rules fail to be evaluated if an RDL expression in the condition or in the variables returns a null error, i. e. if the last evaluated operator in the expression's tree returns a null error. In that case Pulse skips that rule and continues the workflow with the next rule or the next workflow element.

Check the **Input parameters** column in the following table to see whether the respective operator returns a null exception when receiving a null value in the input value.

In the case of operations over collections, check the **Description** column of the table to see which ones return a null exception when the collection is `null`.

Comparison operators

`==`, `!=`

Description	Input parameters	Examples
Checks if the two operands are equal or not.	Any primitive data type . The two operands must be of the same type.	<pre>card_present_flag == true</pre>

Description	Input parameters	Examples
		<pre>merchant_country == 'EU'</pre>

<, >, <=, >=

Description	Input parameters	Examples
Compares two values of an ordered domain.	Numeric data types: <code>int</code> , <code>long</code> , <code>float</code> , <code>double</code>	<pre>amount > average_amount_per_card_in_1_month * 1.5</pre>

Arithmetic operators

+, -, *, /, %

Description	Input parameters	Examples
The arithmetic operators obey the following type promotion rules: 1. If at least one of the arguments is a double, the result is a double. 2. If at least one of the arguments is a float, the result is a float, unless the result is a double because of the first rule. 3. If at least one of the arguments is a long, the result is a long, unless the result is a double or float because of the first two rules.	Numeric types: <code>int</code>, <code>long</code>, <code>float</code>, <code>double</code> The arithmetic operators throw a null exception if any of the input parameters is <code>null</code>.	<pre>5%2</pre> Returns the integer remainder of the division 5/2.

Boolean operators

and

Description	Input parameters	Examples
Returns <code>true</code> if both received arguments are true.	Boolean The and operator throws a null exception in the following cases: • <code>nullExpression</code> and <code>true</code> • <code>nullExpression</code> and <code>false</code> • <code>true</code> and <code>nullExpression</code>	<pre>card_present_flag == false and merchant_country == 'RU'</pre>

or

Description	Input parameters	Examples
Returns <code>true</code> if at least one of the received arguments is true.	Boolean	<pre>CardWatchList.contains(tokenized_card_PAN) or RiskyCountries.contains(card_country)</pre>

!

Description	Input parameters	Examples
Inverts the value of a boolean expression.	Boolean The <code>!</code> operator throws a null exception if the input parameter is <code>null</code> .	<pre>! CardWatchList.contains(tokeni zed_card_PAN)</pre>

Functions on RDL Maps, and Pulse Lists¹

Lists created in Pulse under **Lists** are represented in RDL as RDL Maps.

MyList [*my_key*]

Description	Input parameters	Examples
Returns the value stored in the List under the referred key.	<i>my_key</i> : the same type as the key of the list	<pre>transactionContinent = Countries[country_code].continent_code;</pre> Fetches the continent code for the current <i>country_code</i> from the <i>Countries</i> list.

MyList.contains(my_key);

Description	Input parameters	Examples
Checks if the List contains a value for a specific key.	<i>my_key</i> : the same type as the key of the list	<pre>isInBlackList = blackList.contains(tokenized_card_PAN) ;</pre>

RDL Lists and RDL Sets¹

Note that RDL Lists are not the same as Lists created in Pulse under **Lists** .

MyCollection [*condition*]

Description	Input parameters	Examples
Creates a window of the collection according to the condition. If the condition defines a single element of a collection, the [] can be used to access the data in that single element. Example: <pre>countryList[country_code].continent_c ode</pre>	condition: see examples on the right	- Create a new list with the countries where the current card (<i>card_number</i>) is whitelisted: <pre>myWhiteListCountries = WhiteList[card_number] .count ry_code;</pre> - Create a new list (aka. window) with the last 5 events that have the same card number as the current event: <pre>lastEvents = event [5 tuples by card_number];</pre> - Access the second event in the lastEvents window: <pre>secondEvent = lastEvents[2];</pre> - Create a new list with the last five events where the terminal type was X or Y:

Description	Input parameters	Examples
		<pre>lastEventsWithTerminalTypeXZ = event [5 tuples where terminal_type == "X" or terminal_type == "Y"];</pre> • Create a new list (aka. window) with the last 3 events of the last 24 hours that have the same card number as the current event: <pre>lastEvents24Hours = event [3 tuples by card_number for 24h];</pre>

MyCollection.contains(expression)

Description	Input parameters	Examples
Checks if the collection contains a specific element.	expression: must match the data type of the collection elements	<pre>cardWatchList.contains(tokeni zed_card_PAN)</pre>

MyCollection.filter(my_field => condition)

Description	Input parameters	Examples
Filters the elements of a collection based on a condition. The operator throws a null exception if the collection is <code>null</code>.	• <i>my_field</i>: a field of the collection • condition: boolean expression	<pre>lastEvents.filter(amount => amount > 200)</pre> Returns only the events where the amount is over 200.

Explicitly iterating over Lists is not recommended

It is not recommended to iterate explicitly over Pulse Lists (e.g., using filter, all, etc), as this is highly inefficient. Pulse Lists are optimized for contains/lookup operations.

MyCollection.any(my_field => condition)

Description	Input parameters	Examples
Returns <code>true</code> if at least one element of the collection fulfills the given condition. The operator throws a null exception if the collection is <code>null</code>. Use it carefully on large collections, as the rule will have to do a check on all incoming transactions, against all list elements.	• <i>my_field</i>: a field of the collection • condition: boolean expression	<pre>myWhiteListCountries.any(country => Countries[country].continent_code == transaction_continent)</pre> Returns <code>true</code> if the <i>myWhiteListCountries</i> list contains any country that is on the same continent as the current event.

Explicitly iterating over Lists is not recommended

It is not recommended to iterate explicitly over Pulse Lists (e.g., using filter, all, etc), as this is highly inefficient. Pulse Lists are optimized for contains/lookup operations.

MyCollection.all(my_field => condition)

Description	Input parameters	Examples
	• <i>my_field</i>: a field of the collection	

Description	Input parameters	Examples
Returns <code>true</code> if all elements of the collection fulfill the condition. Use it carefully on large collections, as the rule will have to do a check on all incoming transactions, against all collection elements. The operator throws a null exception if the collection is <code>null</code>.	condition: boolean expression	<pre>customerDevices.all(device -> device.score < 500)</pre>

Explicitly iterating over Lists is not recommended

It is not recommended to iterate explicitly over Pulse Lists (e.g., using filter, all, etc), as this is highly inefficient. Pulse Lists are optimized for contains/lookup operations.

MyCollection.sum(my_field)

Description	Input parameters	Examples
Sums all the numeric values in the collection. The operator throws a null exception if the collection is <code>null</code>.	<i>my_field</i>: a field of the collection	<pre>lastEvents.sum(amount)</pre>

MyCollection.count()

Description	Input parameters	Examples
Counts the elements in the collection. The operator throws a null exception if the collection is <code>null</code>.	N/A	<pre>lastEvents.count()</pre>

MyCollection.min(my_field)

Description	Input parameters	Examples
Returns the lowest element in the collection. The operator throws a null exception if the collection is <code>null</code>.	<i>my_field</i>: a field of the collection	<pre>lastEvents.min(amount)</pre>

MyCollection.max(my_field)

Description	Input parameters	Examples
Returns the highest element in the collection. The operator throws a null exception if the collection is <code>null</code>.	<i>my_field</i>: a field of the collection	<pre>lastEvents.max(amount)</pre>

MyCollection.avg(my_field)

Description	Input parameters	Examples
Returns the average of the elements in the collection. The operator throws a null exception if the collection is <code>null</code>.	<i>my_field</i>: a field of the collection	<pre>lastEvents.avg(amount)</pre>

RDL Lists

Note that RDL Lists are **not** the same as Lists created in Pulse under **Lists**. See the previous section.

MyRDList.at(my_index)

Description	Input parameters	Examples
Applies only to RDL Lists. Returns the value stored in the RDL List under the referred index. The at operator throws a null exception if the input parameter is <code>null</code>.	<i>my_index</i>: integer	<pre>lastEvents.at(2)</pre> Returns the second event of the <i>lastEvents</i> list.

Selecting a value dynamically

outputVariable = **match** if condition then expression1; else expression2; end end

Description	Input parameters	Examples
Implements an if-then-else block. If the condition is true, the output variable receives the value of expression1. Otherwise the output variable receives the value of expression2.	<i>outputVariable</i>: any type. - condition: boolean expression. - expression1 and expression2: the same data type as the <i>outputVariable</i>.	<pre>events = event [3 tuples by card_number for 24h]; prevEvent = match if events.count() > 2 then return events.at(2); else if events. count() > 1 then return events.at(1); else return null; end end Returns the last event from the <i>events</i> list.</pre>

Others

expression1 ?? expression2

Description	Input parameters	Examples
The coalesce (??) operator returns expression1, unless expression1 is <code>null</code>, in which case it returns expression2. In RDL, the coalesce operator has the highest precedence when used in conjunction with arithmetic or boolean operators, meaning that it gets evaluated before the other operators.	Any	Return -1 if <i>MyList</i> is empty, otherwise return the amount average of the list: <pre>MyList.avg(amount) ?? -1</pre>

Description	Input parameters	Examples
		Example to illustrate the precedence in relation to other operators: <pre>1 + null ?? 3 + 4</pre> This expression is equivalent to <pre>1 + (null ?? 3) + 4</pre> ...and returns 8 as the result.

UDF (User-defined function) calls

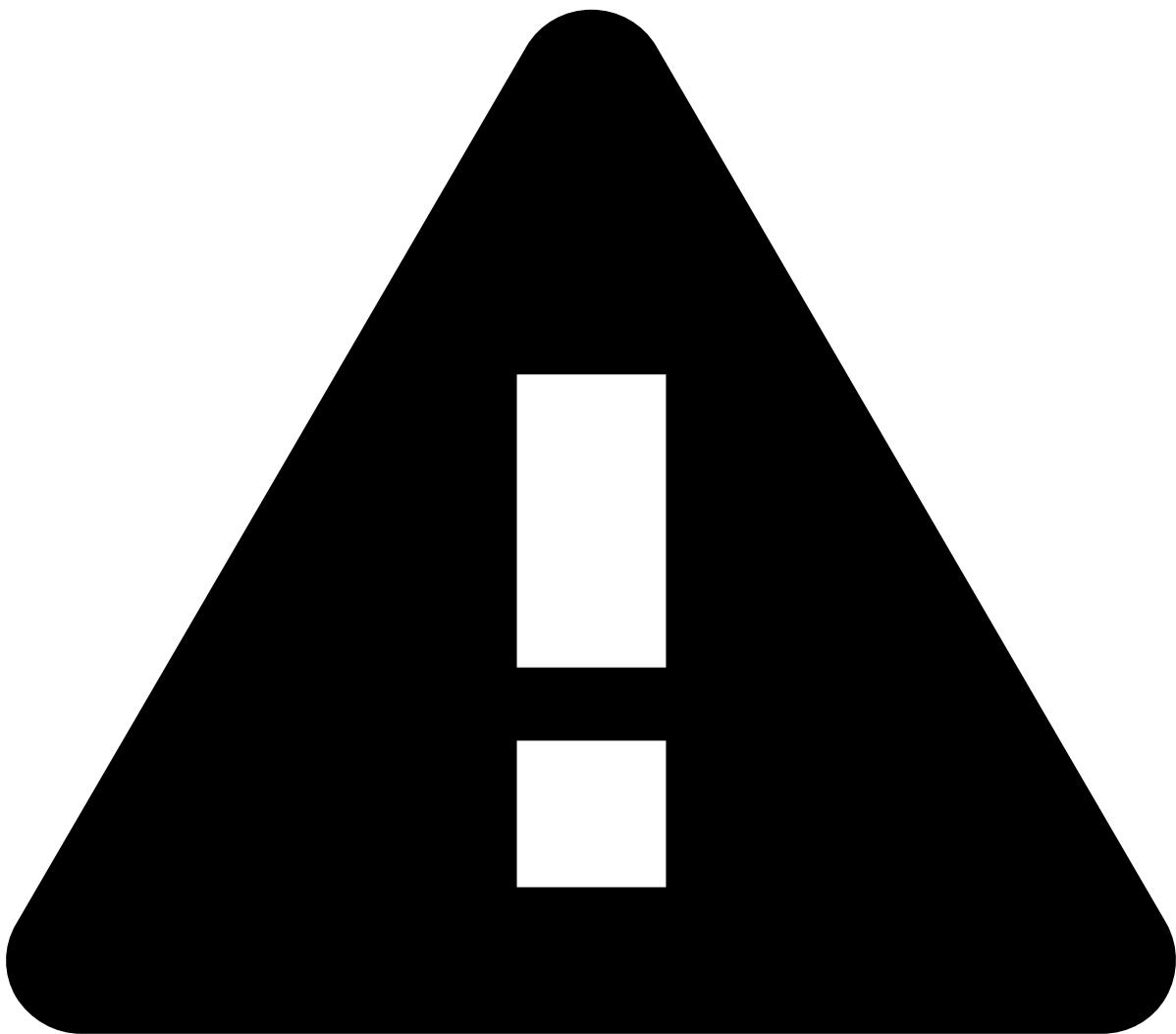
Description	Input parameters	Examples
You can call User-Defined Functions to perform specific computations.		- Find the location of the substring "775" in the <i>card_acceptor_id</i>: <pre>matchLocation = StringUtils.indexOf(card_acceptor_id , "775");</pre> - Fetch the first letter 3 positions after the match location found with the above query: <pre>letterAfterMatch = StringUtils.substring(card_acceptor_id);</pre> - Check if the email_domain field matches the defined Java pattern: <pre>isNumericDomain = StringTools.regexMatches("[0-9]+.com", email_domain);</pre> - From the complex field called <i>countries_map</i>, retrieves the string value associated to the "USA" key: <pre>CollectionUDFs.getString(countries_map, "USA")</pre> - From the complex field called <i>countries_map</i>, retrieves the integer value associated to the "USA" key: <pre>CollectionUDFs.getInt(call_code_map, "USA")</pre>

Operator precedence

The following table lists RDL operators according to their precedence order. Operators at the top of the table have higher precedence.

Operator	Precedence
grouping	<code>(expr)</code>
unary	<code>+expr</code> <code>-expr</code> <code>!</code>
cast	<code>(type) expr</code>
coalesce	

Operator	Precedence
	?? Note that this operator has a different precedence in PQL.
arithmetic – multiplicative	* / %
arithmetic – additive	+ -
comparison	> >= == <= < !=
logical – OR	or
logical – AND	and
assignment	=



warning

Unlike other languages, in RDL the operator `or` takes precedence over the `and` operator.

The following rules apply:

- Operators with higher precedence are evaluated before the ones with lower precedence.
- Operators in the same line have the same precedence:
 - All binary operators except for the assignment operator are evaluated from left to right.
 - Assignment operator is evaluated right to left.